

Statement 1

Answer: True, it only rotates about the y-axis, so the kinetic energy of the body should only be influenced by the angular rotation rate and moment of the inertia about the y axis.

$$\text{Such that } T = \frac{1}{2} I_y \omega^2$$

Statement 2.

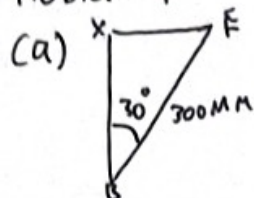
Answer: True, the equation of friction force is $F = \mu_k N$.
Where μ_k is the kinetic friction coefficient and N is the normal force acting from the surface to the drum.

Statement 3

Answer: True, in an ideal situation, the total kinetic energy of the system will stay the same.

If however, there is air friction, or the possibility of the energy transforming into thermal or sound energy, the total kinetic energy of the system will reduce.

Problem 1



$$XB = 0.3 \cos 30 = 0.26 \text{ m}$$

$$Mgh = \frac{1}{2} Mv^2$$

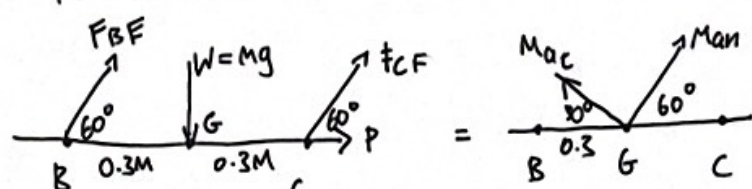
$$v = \sqrt{2gh}$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.26}$$

$$= 2.26 \text{ m/s (Velocity of B and D)}$$

Type of motion for rod ABCD: Translation

(b) FBD and motion diagram:



For the link:

$\omega = 6 \text{ rad/s}$
 $\alpha = 12 \text{ rad/s}^2$

(d) $a_n = r\omega^2$
 $= 0.3 \cdot 6^2 = 10.8 \text{ rad/s}^2$

$a_t = r\alpha$
 $= 0.3 \cdot 12 = 3.6 \text{ rad/s}^2$

$\sum M_B = \sum (M_B)_{\text{eff}}$

$$F_{CF} \sin 60 (0.6) - W(0.3) = M a_t \sin 30 (0.3) + M a_n \sin 60 (0.3)$$

$$0.52 F_{CF} - 6 \cdot (9.81)(0.3) = 6(0.3) [3.6 \sin 30 + 10.8 \sin 60]$$

$$0.52 F_{CF} = 37.73$$

$$\underline{\underline{F_{CF} = 72.56 \text{ N} = F_{BE}}}$$

(c) $\sum F_x$:

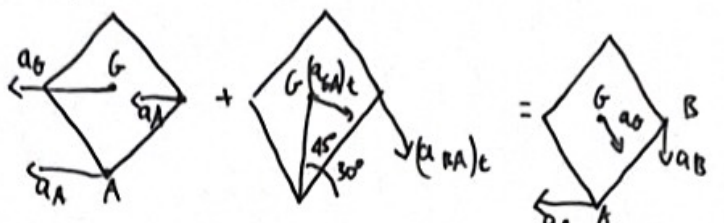
$$(F_{BE} + F_{CF}) \cos 60 + P = -M a_t \cos 30 + M a_n \cos 60$$

$$72.56 + P = M(a_n \cos 60 - a_t \cos 30)$$

$$P = 6(10.8 \cos 60 - 3.6 \cos 30) - 72.56$$

$$= -58.87 \text{ N or } 58.87 \text{ N (←)}$$

Problem 2



$$(a_{GA})\epsilon = \alpha r_{GA}$$

$$= \alpha \cdot \frac{\sqrt{0.15^2 + 0.15^2}}{2}$$

$$= 0.1061\alpha$$

$$(a_{BA})\epsilon = \alpha r_{BA}$$

$$= 0.15\alpha$$

From the x component: $a_B = a_A + a_{BA}$

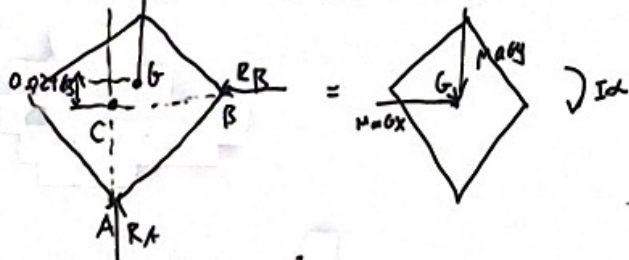
$$0 = -a_A + 0.15\alpha \sin 45^\circ$$

$$a_A = 0.075\alpha$$

$$a_G = a_A + a_{GA}$$

$$= (-0.075\alpha i) + [(a_{GA})\epsilon \cos 45^\circ - (a_{GA})\epsilon \sin 45^\circ j]$$

$$= 0.02745\alpha i - 0.02745\alpha j$$



$$I = \frac{1}{12} M (0.15^2 + 0.15^2)$$

$$= \frac{1}{12} (2.5) (0.15^2 + 0.15^2) = 0.0094$$

$$M a_{Gx} = 2.5(0.02745\alpha) = 0.069\alpha$$

$$M a_{Gy} = 2.5(0.02745\alpha) = 0.069\alpha$$

(a) Moment at P:

$$M g r_{GC} = I\alpha + 2 [(0.069\alpha)(0.02745)]$$

$$2.5(9.81)(0.02745) = 0.0094\alpha + 2(1.89 \cdot 10^{-3}\alpha)$$

$$0.673 = 0.01318\alpha$$

$$\alpha = 51.1 \text{ rad/s}^2 \text{ (clockwise)}$$

(b) $\sum F_y = M a_y$

$$R_A - Mg = -0.069(51.1)$$

$$R_A = -0.069(51.1) + 2.5(9.81)$$

$$= 21 \text{ N } (\uparrow \text{ upwards})$$

Problem 3

Conservation of energy:

$$T_1 + V_1 = T_2 + V_2$$

Initial kinetic energy: $T_1 = \frac{1}{2} M_C (V_{r1})^2$

Initial momentum: $V_1 = \frac{1}{2} I_1 (\omega_1)^2$

End kinetic energy: $T_2 = 0$

End momentum: $V_2 = \frac{1}{2} I_2 \omega_2^2$

$$\frac{1}{2} M_C (V_{r1})^2 + \frac{1}{2} I_1 (\omega_1)^2 = \frac{1}{2} I_2 \omega_2^2$$

$$M_C (V_{r1})^2 = I_2 (\omega_2)^2 - I_1 (\omega_1)^2$$

Because, $I_1 \omega_1 = I_2 \omega_2$ (conservation of angular momentum)

$$\text{so, } \omega_2 = \frac{I_1 \omega_1}{I_2}$$

$$M_C (V_{r1})^2 = I_2 \left(\frac{I_1 \omega_1}{I_2} \right)^2 - I_1 (\omega_1)^2$$

$$\begin{aligned} I_1 &= 1.2 + 8r^2 \\ &= 1.2 + 8 \cdot (0.6)^2 \\ &= 4.08 \end{aligned}$$

$$M_C (V_{r1})^2 = I_2 \left(\frac{I_1 \omega_1}{I_2} \right)^2 - I_1 (\omega_1)^2$$

$$8 \cdot (1.5)^2 = I_2 \left(\frac{I_1}{I_2} - 1 \right) (\omega_1)^2$$

$$\frac{8 \cdot (1.5)^2}{(1.5)^2} = 4.08 \frac{I_1}{I_2} - 4.08$$

$$\frac{8 + 4.08}{4.08} = \frac{I_1}{I_2}$$

$$2.96 = \frac{4.08}{I_2}$$

$$I_2 = 1.378$$

(a) $I = I_2 = 1.2 + 8r_2^2$

$$1.378 = 1.2 + 8(r_2)^2$$

$$\underline{r_2 = 0.15 \text{ M (minimum distance)}}$$

(b) $\omega_2 = \frac{I_1 \omega_1}{I_2}$

$$= \frac{4.08 \cdot 1.5}{1.378}$$

$$= \underline{\underline{4.44 \text{ rad/s}}}$$

Problem 4

$$(a) mgh = \frac{1}{2}mv_1^2$$

$$v_1^2 = 2gh$$

$$v_1 = \sqrt{2gh}$$

$$= \sqrt{2 \cdot 9.81 \cdot 2.5}$$

$$= \underline{\underline{7 \text{ M/s}}} \text{ (velocity of gymnast B)}$$

(b)

Moment at D:

$$M_B V_1 L = \frac{1}{12} M_P (2L)^2 \omega_2 + (M_A + M_B)(L\omega_2)L$$

$$M_B V_1 = \frac{1}{3} M_P L \omega_2 + (M_A + M_B) L \omega_2$$

$$M_B V_1 = \left(\frac{1}{3} M_P + M_A + M_B \right) L \omega_2$$

$$\omega_2 = \frac{M_B V_1}{\left(\frac{1}{3} M_P + M_A + M_B \right) L}$$

$$V_2 = L \omega_2$$

$$= \cancel{L} \cdot \frac{M_B V_1}{\left(\frac{1}{3} M_P + M_A + M_B \right) \cancel{L}}$$

$$= \frac{70 \cdot 7}{\left(\frac{1}{3} \cdot 15 + 55 + 70 \right)}$$

$$= \underline{\underline{3.769 \text{ M/s}}} \text{ (velocity of gymnast A)}$$

$$(c) \frac{1}{2}mv_2^2 = mgh$$

$$h = \frac{v_2^2}{2g}$$

$$= \frac{(3.769)^2}{2 \cdot (9.81)}$$

$$= 0.724 \text{ M}$$